

B. NIAH Setting

We largely follows the passkey retrieval prompt template of Mohtashami and Jaggi (2023) but using 7-digit passkey and Paul Graham Essays² as the background filler, as set forth in Arize-ai and Reid et al. (2024):

There is an important info hidden inside a lot of irrelevant text. Find it and memorize them. I will quiz you about the important information there.

<prefix filled by Paul Graham Essays>

The pass key is <7-DIGIT PASS KEY>. Remember it. <7-DIGIT PASS KEY> is the pass key.

<suffix filler>

What is the pass key? The pass key is

C. More Ablation Results

In our efficiency evaluation, we observe that with a residual length of 32, KIVI achieves a significantly higher memory compression rate, which in turn leads to increased throughput. Additionally, our ablation study reveals that changing the residual length from 128 to 32 does not result in a substantial performance gap. We demonstrate KIVI with a residual length of 32 across all benchmark datasets. As shown in Tables 6 and 7, KIVI with a residual length of 32 also delivers performance comparable to that of the 16-bit full model.

Table 6: Performance comparison between 16bit, KIVI-2 (2bit) / KIVI-4 (4bit) with residual length 128 and 32 across various models. R32 stands for residual length 32.

Model		CoQA	TruthfulQA	GSM8K
Llama-2-7B	16bit	63.88	30.76	13.50
	KIVI-2 R128	63.05	33.95	12.74
	KIVI-2 R32	62.85	33.01	13.57
Llama-2-13B	16bit	66.37	29.53	22.67
	KIVI-2 R128	66.23	29.84	20.77
	KIVI-2 R32	66.57	29.35	20.62
Falcon-7B	16bit	59.83	23.20	4.55
	KIVI-4 R128	59.67	22.58	4.47
	KIVI-4 R32	59.73	22.96	3.94
	KIVI-2 R128	57.48	24.98	3.41
	KIVI-2 R32	57.50	25.70	2.20
Mistral-7B	16bit	67.40	30.45	38.36
	KIVI-2 R128	66.35	32.17	36.01
	KIVI-2 R32	65.90	31.21	34.34

D. More Experimental Results

We present additional results using Llama3-8B, Mistral-7B-v0.2, and LongChat-7B-v1.5 in LongBench, which can be found in Table 8, Table 9 and Table 10, respectively.

We also show result of Needle-in-a-Haystack Test in Figure 4. The settings largely follow the format of the original passkey retrieval task (Mohtashami and Jaggi, 2023) while including some modern modifications set forward by Arize-ai and the technical report of Gemini 1.5 (Reid et al., 2024).

²<https://paulgraham.com/articles.html>

Table 7: Performance evaluation of KIVI with residual length 128 and 32 on various models across a range of benchmarks in LongBench. R32 stands for residual length 32.

Model		Qasper	QMSum	MultiNews	TREC	TriviaQA	SAMSum	LCC	RepoBench-P	Average
Llama2-7B	16bit	9.52	21.28	3.51	66.00	87.72	41.69	66.66	59.82	44.52
	KIVI-2 R128	9.31	20.50	1.14	66.00	87.42	42.71	66.88	60.23	44.27
	KIVI-2 R32	9.26	20.53	0.97	66.00	87.42	42.61	66.22	59.67	44.08
Llama2-13B	16bit	9.32	21.38	3.71	70.00	87.87	43.55	66.61	56.42	44.85
	KIVI-2 R128	8.58	20.69	6.19	69.50	87.78	44.30	65.08	55.46	44.69
	KIVI-2 R32	8.38	20.74	7.01	69.50	87.78	44.43	64.89	55.31	44.75
Llama2-7B-Chat	16bit	19.65	20.54	26.36	63.00	84.28	41.12	59.75	52.93	45.95
	KIVI-2 R128	19.32	20.46	25.48	63.00	84.84	40.60	58.71	52.97	45.67
	KIVI-2 R32	19.10	20.08	25.33	63.00	85.04	39.80	57.91	52.38	45.33
Llama2-13B-Chat	16bit	24.18	20.37	25.69	67.50	86.90	42.18	50.23	50.64	45.96
	KIVI-2 R128	23.59	20.76	25.25	67.50	87.17	41.56	49.93	48.45	45.52
	KIVI-2 R32	23.56	20.90	25.45	67.50	87.42	41.40	48.93	48.81	45.49
Falcon-7B	16bit	1.48	2.35	11.09	13.00	5.84	2.44	23.86	9.69	8.71
	KIVI-4 R128	1.04	2.41	11.98	13.00	5.84	2.36	23.72	9.92	8.78
	KIVI-4 R32	1.03	2.45	11.99	13.50	5.84	2.46	23.88	9.95	8.88
	KIVI-2 R128	1.98	3.61	6.78	10.00	6.24	2.73	22.18	10.12	7.95
	KIVI-2 R32	2.28	3.23	6.73	10.00	6.31	2.88	22.71	10.45	8.07
Mistral-7B	16bit	8.12	19.98	19.99	67.50	89.80	41.69	66.59	58.99	46.58
	KIVI-2 R128	6.92	19.71	17.92	66.50	89.63	41.66	65.52	58.99	45.85
	KIVI-2 R32	6.84	19.81	17.20	66.50	89.63	42.82	65.13	58.06	45.74

Table 8: The results of Llama-3-8B-Instruct with KIVI on LongBench. The model has 8K context length and applies group query attention, which uses 8 heads for KV cache instead of the full 32 heads. We use a 32 group size and 128 residual length for both KIVI-2 and KIVI-4. The baseline is of full precision.

	NarrativeQA	Qasper	MultiFieldQA	HotpotQA	MuSiQue	2WikiMQA	GovReport	QMSum
Baseline	21.71	44.24	44.54	46.82	21.49	36.42	30.03	22.67
w./ KIVI-2	21.35	43.17	44.49	46.79	20.56	37.05	29.98	22.07
w./ KIVI-4	21.01	44.83	44.60	46.96	21.43	36.48	30.22	22.44
	MultiNews	LCC	RepoBench-P	TriviaQA	SAMSum	TRec	PR	Avg
Baseline	27.79	57.00	51.22	90.23	42.53	74.50	67.00	45.21
w./ KIVI-2	27.77	50.84	46.65	90.54	42.26	74.50	67.50	44.37
w./ KIVI-4	27.97	57.36	52.03	90.33	42.97	74.50	66.50	45.31

Table 9: The results of Mistral-7B-Instruct-v0.2 with KIVI on LongBench. The model has 32K context length and applies group query attention, which uses 8 heads for KV cache instead of the full 32 heads. We use a 32 group size and 128 residual length for both KIVI-2 and KIVI-4. The baseline is of full precision.

	NarrativeQA	Qasper	MultiFieldQA	HotpotQA	MuSiQue	2WikiMQA	GovReport	QMSum
Baseline	21.02	29.41	47.13	36.53	19.13	21.76	32.59	23.99
w./ KIVI-2	20.61	28.73	44.88	35.47	17.95	20.68	32.55	23.65
w./ KIVI-4	20.97	29.41	46.52	36.25	19.53	21.66	32.97	24.06
	MultiNews	LCC	RepoBench-P	TriviaQA	SAMSum	TRec	PR	Avg
Baseline	27.09	53.49	51.40	86.23	43.04	71.00	89.33	43.54
w./ KIVI-2	26.54	53.03	51.16	86.00	43.34	71.00	80.83	42.43
w./ KIVI-4	26.89	53.33	51.41	86.23	43.34	71.00	89.42	43.53

Table 10: The results of LongChat-7B-v1.5-32K with KIVI on LongBench. The model has 32K context length. We use a 32 group size and 128 residual length for both KIVI-2 and KIVI-4. The baseline is of full precision.

	NarrativeQA	Qasper	MultiFieldQA	HotpotQA	MuSiQue	2WikiMQA	GovReport	QMSum
Baseline	20.65	29.42	43.15	33.05	14.66	24.14	30.85	22.84
w./ KIVI-2	20.79	28.69	41.02	32.91	13.82	23.00	30.47	22.59
w./ KIVI-4	20.49	28.90	43.24	33.07	14.66	24.86	31.40	22.84
	MultiNews	LCC	RepoBench-P	TriviaQA	SAMSum	TRec	PR	Avg
Baseline	26.55	54.83	58.94	83.99	40.75	66.50	30.50	38.72
w./ KIVI-2	26.28	54.11	57.62	83.19	41.28	66.50	32.25	38.30
w./ KIVI-4	26.52	54.06	58.77	83.88	40.62	67.00	31.50	38.79